VENKATESH MALLARAPU

Data Scientist | ML Engineer | MLOps
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SUMMARY

B.Tech Computer Science graduate and aspiring Data Scientist / ML Engineer with hands-on experience in Python, SQL, machine learning and MLOps. Built end-to-end projects in fraud detection, AI agents and computer vision, from model training to REST API deployment with CI/CD, and completed a Data Scientist internship on demand forecasting.

EDUCATION

Holy Mary Institute of Technology & Science, Hyderabad, India
Bachelor of Technology in Computer Science **November 2022 - August 2026**

SKILLS SUMMARY

- **Languages:** Python, SQL
- **ML & Gen AI:** Regression, Classification, Clustering, Random Forest, SVM, KNN, Naive Bayes, XGBoost, LSTM, Decision Trees, Prompt Engineering, RAG Pipelines, AI Agents, LLM Integration (OpenAI/HuggingFace)
- **Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Flask, Matplotlib, Seaborn
- **MLOps & Governance:** CI/CD for ML (GitHub Actions), Model Versioning, REST API Model Serving, Lineage Tracking, Drift Detection, Model Monitoring
- **Tools:** Jupyter Notebook, Google Colab, Git/GitHub, VS Code, Elasticsearch, Splunk
- **Data Analysis:** EDA, Feature Engineering, Data Cleaning, SMOTE, Outlier Handling
- **CS Fundamentals:** Data Structures & Algorithms (Trees, Graphs, Hash Maps, DP, BFS/DFS)

WORK EXPERIENCE

DATA SCIENTIST INTERN | Evoastra Ventures **February 2026 - April 2026**

- Performed EDA, preprocessing and feature engineering on real-world business datasets, uncovering key patterns that shaped model design.
- Built Random Forest, SVM and Regression models for supply chain optimization and demand forecasting, improving prediction accuracy by ~15%.
- Deployed models via REST APIs for real-time inference and implemented CI/CD steps for reproducible, versioned ML deployments.
- Integrated model monitoring and governance tools to track post-deployment performance and maintained lineage documentation across experiments.
- Created Matplotlib and Seaborn dashboards that communicated actionable insights to stakeholders.

PROJECTS

Credit Card Fraud Detection | LINK

- Built a real-time fraud classifier on highly imbalanced data (fraud < 0.2%) using StandardScaler on PCA features and SMOTE oversampling.
- Benchmarked Logistic Regression, XGBoost and Isolation Forest against a 200-estimator Random Forest, achieving 0.95+ PR-AUC on the fraud class.
- Built a CI/CD pipeline for automated retraining with lineage tracking; served the model via a Flask REST API to a live fraud-monitoring dashboard.

AI Company Intelligence Agent | LINK

- Architected an autonomous ReAct-style AI agent for company research, orchestrating web search, scraping, financial extraction and report generation.
- Implemented a reason-act loop in which the agent selects tools (search, fetch, parse) and feeds results back into context for the next step.
- Built a Node.js/Express layer for session state, tool routing and Server-Sent Events streaming of real-time company intelligence cards.

Handwritten Character Recognition | LINK

- Trained a deep CNN on EMNIST Balanced (814,255 images, 62 classes) with Batch Normalization and Dropout, reaching ~92% top-1 accuracy and sub-100ms inference.
- Used Keras augmentation and OpenCV preprocessing (grayscale, Gaussian blur, adaptive thresholding, contour bounding boxes) to improve generalization.
- Deployed a Flask REST API that serves real-time predictions with confidence scores to an HTML5 Canvas frontend.

CERTIFICATES

- **Machine Learning, Python, Artificial Intelligence** | Visualpath
- **Data Science** | Tutedude